

Observer Patch Holography as a Strange-Loop Self-Simulation

The Strongest Results and Their Certificates

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Observer-Patch Holography (OPH) is a theory-of-everything candidate with declared structural and conditional theorem branches, arithmetic fixed-point certificates, and public falsifiers. Its 111 sorry-free Lean theorems cover a declared finite subset, not physical branch entry, the A_5 port-frame and classification steps, trace-balanced global form, or physical descent. Quantum mechanics, general relativity, and Standard Model structure appear on their named receipt branches. Numerical map roots have no physical status without their transport, scale, pole, and provenance gates.

OPH begins with a self-referential mathematical structure with no external machine or clock. Its framework additionally declares observer-like systems: each observer is a bounded self-reading patch with local state, ports, boundary readback, durable records, mismatch feedback, repair moves, and public receipts, and no patch contains the world. The agreement condition defines a stable public fixed point,

$$T(\mathfrak{U}_{\text{OPH}}) = \mathfrak{U}_{\text{OPH}}.$$

The stack gives Born–Lüders probabilities on the declared central record algebra [O1,O3], receipt-conditional Lorentzian and Einstein branches [O4,C8], and the realized MAR/tensor Standard Model branch with three colors, three generations, and one Higgs doublet [O4]. The de Sitter capacity closure is conditional on its readback map and premises. The fine-structure roots certify incomplete maps, not physical endpoints.

How to read the argument

This document runs one assent chain. **Step 1:** grant the six strange-loop principles below [C17]. **Step 2:** verify the certificates; the interval uniqueness certificate for the pixel closure [C1], the conditional capacity fixed point [C16], and the 111 sorry-free Lean theorems [C6] are machine-checkable objects on their stated scopes. **Step 3:** observe the landings in the results table, each carrying its epistemic label. What survives as freedom after these three steps is the enumerated open program: the Ward-projected hadronic transport term, the electroweak physical readout contract, and the capacity readback map F with its premises. Each has a declared completion criterion. Only claim surfaces with complete statements, stable hypotheses, and fixed comparison objects enter the public falsification program. A reader can therefore disagree at a specific point: a principle, a certificate, a branch receipt, or the status assigned in the closure ledger.

The strange-loop principles [C17]

The conditional chain uses six declared strange-loop principles, condensed from the program’s claim registry [C17]. They are premises, not consequences of SL-0. **SL-0 (Strange loop):** reality is a self-referential mathematical structure that explains itself, the fixed point of a self-reading computation; the simulating structure and the simulated world are one closed system, with no external machine and no external clock. Observer agreement defines the public fixed-point target. **SL-1 (Substrate chart):** the observer-facing substrate is charted by the two-sphere with the screen-microphysics architecture (patches, records, overlap comparison, repair); 3 + 1-dimensional observer experience is a design constraint of the substrate under SL-0. **SL-2 (Detuning law):** the pixel ratio sits off the golden-ratio balance point by the Gaussian-normalized width of one electromagnetic observation, $P = \varphi + \sqrt{\pi} \alpha$; a screen at exact balance carries no events. **SL-3 (Constant identity):** under self-simulation the fine-structure constant of the simulated world and the substrate pixel readout are one quantity; the theory computes P from its closure equation and the measured α is the test that value faces. **SL-4 (Capacity identity):** the total record capacity N is one number shared by substrate and world; the theory computes N from $N = F(N)$ and the Λ readout is the test that value faces. **SL-5 (SI bridge):** one clock anchor, the cesium transition, connects substrate units to laboratory units.

Zero dials

The theory layer takes no measured number: zero continuous fit parameters in every defining equation. Both constants are defined by closure equations,

$$P_\star = \Gamma_P(P_\star), \quad N_\star = \Gamma_N(N_\star),$$

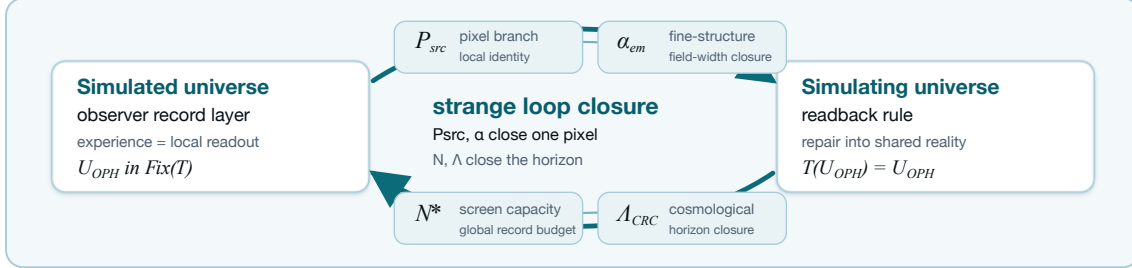
and nothing in the architecture can be adjusted toward data. Discrete structural selections are counted, with their menu sizes, in the public closure ledger [C17]. Measured values enter at the test layer as comparisons, and at the working layer as located values inside lanes whose closure terms are open: the working pixel $P = 1.630968209403959 \dots$ located from CODATA $\alpha^{-1} = 137.035999177(21)$ [S2], and the working capacity $N = 3.31 \times 10^{122}$ located from the Λ readout [S5]. The capacity borrow is the strange loop operating as designed: in a self-simulation, measurement and substrate readback are one act, so the universe’s own Λ readout is the legitimate instrument that locates N while the forcing equation is work in progress. Every working borrow is a located input inside a lane whose generator is work in progress. The integers $(N_c, N_g, \beta_{\text{EW}}) = (3, 3, 4)$ are branch outputs, and their alternatives are structurally excluded by the declared enumeration.

Exactly one fixed point, certified

The pixel closure has exactly one fixed point per declared map on the declared physical domain $\alpha^{-1} \in [100, 200]$: existence and local uniqueness by interval Banach in outward-rounded arithmetic with contraction constant $L \leq 0.0724$, and global at-most-one by certified $\sup |g'| < 1$ on all 256 subdivision pieces with zero exceptional set [C1]. The enclosure width is 7.2×10^{-24} in α^{-1} , with the SU(2)/SU(3) edge-sum tails bounded by geometric majorants so the enclosure covers the infinite-cutoff sums. An adversarial third-party audit reproduced the certificate arithmetic end to end. A second fixed point on the declared physical domain is arithmetically excluded. The maximal-domain extension certificate gives the same result on the full analytic domain of each declared map, with an empty exclusion list [C1]. This is uniqueness of the stated maps. It does not complete their physical transport.

The flagship numbers

The gauge-width declared map lands at $\alpha^{-1} = 137.035660$, a relative 2.5×10^{-6} from the measured $\alpha^{-1} = 137.035999177(21)$ [S2], a certified fixed point with the Ward-projected hadronic transport term open. The source map lands at 136.994835, a relative 3.0×10^{-4} , under the same label. The chart W coordinate sits 0.5 propagated experimental standard deviations from the PDG 2026 complex pole [S3], a convention diagnostic with the physical readout contract open. Read as $\Lambda \ell_P^2 = 3e^{-6\pi/(P\alpha_U)}$, the capacity bridge writes the 122-order vacuum-energy hierarchy as one exponential of the inverse gauge coupling, with zero continuous freedom once α is measured and no 10^{120} fine-tuning anywhere. The closure equation that forces the measured capacity exactly is work in progress: the candidate bridge sits 6.6% from the Λ -located capacity [S5], conditional on the readback map F and the premises CP-1-CP-3. The structural stack delivers Lorentz kinematics, the Einstein branch, the Standard Model quotient, three generations, and one Higgs on their declared branches. The 111-theorem Lean artifact [C6] covers its declared finite subset.



The selection chain and the uniqueness endgame

The claim lattice has one logical shape, stated in full in the consistency stack [C15]: ten consistency requirements C1-C10 quotient the space of self-consistent worlds. Self-reading forces records and repair (C1); overlap consensus forces the public quotient, and on the S^2 chart forces signature, dimension, and the Lorentz group (C2); modular self-consistency forces internal time and thermality (C3); transportable-charge consistency forces the compact gauge menu and, with the declared matter package, the Standard Model quotient (C4); entropic consistency forces the Einstein branch and leaves one global number (C5); record existence forces the detuning law and leaves one local number (C6). After C1-C6 the surviving freedom is the two-real-parameter family (P, N) together with the declared discrete selections the ledger counts. C7 identifies the two parameters with measured readouts; C8-C9 are two equations on the family; C10 removes units. Three lemmas close the chain. **L1**: the pixel closure has exactly one fixed point per readout map on the declared physical domain, by the interval and global uniqueness certificates [C1]. **L2**: the same schema applies to N once the readback map F is constructed; its required properties are given by the readback specification [C16]. **L3**: under L1+L2 the principle set admits at most one (P, N) ; for P this is a theorem on the declared physical domain. If the two equations close and the fixed points are unique, the principle set admits at most one pair (P, N) . Identifying that pair with our universe requires the physical readback and comparison maps recorded as open in the closure ledger. No cosmological target is locked.

Quantitative results beside measurement

Each row carries its evidence class, stated once and in full. Every value is an output of the named declared map or branch, with zero continuous fit parameters.

Quantity	OPH	Reference	Distance	Evidence class
α^{-1} , gauge-width map	137.035660	137.035999177(21) [S2]	2.5×10^{-6} rel.	certified fixed point of the declared map; hadronic transport open
α^{-1} , source map	136.994835	137.035999177(21) [S2]	3.0×10^{-4} rel.	certified fixed point of the declared map; hadronic transport open
Capacity at Λ	3.532×10^{122}	3.313×10^{122} [S5]	6.6% (candidate)	closure equation $N = F(N)$ is work in progress; the exponential hierarchy is structural, the exact bridge is unclosed

The two α rows are landings of the declared maps; the Ward-projected hadronic transport term of the physical Thomson readout is open, so neither row is scored as a physical endpoint verdict. The capacity row is a work-in-progress lane: the structural result is the exponential hierarchy, and the closure equation that would force the measured capacity exactly is the object under search.

Falsification scope

The falsification program is restricted to completed non-cosmological claim surfaces: finite consensus, collar recovery, Lorentz-Einstein reconstruction, compact-gauge and realized Standard-Model structure, declared-action classical carriers, and finite screen sieves. It contains ten mathematical and five physical realized-branch falsifiers. Cosmology, dark matter, quantitative mass/readout closures, black-hole continuations, and engineering lanes are work in progress and carry no falsification verdict.



1. The observer-patch fixed point

The mathematical object is a finite patch system. Patches expose ports, write records, compare overlaps, repair disagreement, and return a shared normal form [O1,O2]. Let T be the observer-overlap repair map and let $\mathcal{U}_{\text{OPH}}$ denote the selected normal form. The simulation identity is

$$T(\mathcal{U}_{\text{OPH}}) = \mathcal{U}_{\text{OPH}}.$$

The OPH simulation theorem establishes this identity from endogenous update, observer-readable records, schedule-independent overlap repair, selected-branch elimination, and implementation/clock closure [O2]. The universe is the stable record world returned by its own readback-and-repair computation.

The concrete carrier is a finite collection of local states, ports, central records, repair maps, and checkpoints. The reference benchmark replays all six overlap schedules and returns the same repaired normal form on the clean abelian branches [C5]. A separate Lean 4/Mathlib artifact certifies observer confluence, approximate schedule independence, refinement naturality, and the contractive conditional-resampling projector, part of the 111 sorry-free theorems that carry the consensus core and the coupling algebra [C6].

Quantum event surface

On a completed compare/write/verify slice, observer-accessible events generate a finite central record algebra. For an event E ,

$$\mathbb{P}(E) = \text{Tr}(\rho P_E), \quad \rho|_E = \frac{P_E \rho P_E}{\text{Tr}(\rho P_E)}.$$

These are the Born rule and Lüders update. On the associated two-wing Bell surface,

$$|S_{\text{CHSH}}| \leq 2\sqrt{2}.$$

A finite 65,536-record run reproduces the event algebra numerically: committed-event weights sum to 1.0, record projectors have zero idempotency defect, and repeat reads are stable [O1,O3,C7].

2. Relativity, (3+1) events, and Einstein dynamics

The screen readout supplies an oriented conformal S^2 . Its connected conformal group is the connected Lorentz group,

$$\text{Conf}^+(S^2) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1).$$

The corresponding observer-frame space is

$$H^3 \cong \frac{\text{SO}^+(3, 1)}{\text{SO}(3)}, \quad \dim H^3 = 6 - 3 = 3.$$

On the declared event branch, repaired records populate a four-dimensional event base of signature $(-+++)$, while H^3 is the fiber of unit timelike observer frames over each event [O4].

The 2π -normalized modular flow of screen caps supplies the observer-relative clock. Half-sided modular inclusions give positive null translations; null tomography assembles their directional charges into a conserved local T_{ab} . Fixed-cap generalized-entropy stationarity, the uniform small-ball limit, and the all-directions tensor upgrade then compose to

$$G_{ab} + \Lambda g_{ab} = 8\pi G_{\text{geom}} \langle T_{ab} \rangle.$$

Every arrow in this chain has a named theorem and a machine-readable receipt type [O4,C8].

Earned simulation receipts at 64k, 128k, and 1M patches

Three canonical runs instantiate bounded self-reading patches and scale from 65,536 patches/1,024 observers through 131,072/32,000 to 1,048,576/64,000 [C7]. The 128k and 1M runs earn observer-local modular time, gauge-covariant 2π -KMS (Kubo–Martin–Schwinger) replay, a conformal H^3 chart, integer 3 + 1 observer-facing experience, and an emergent agreement field; at 1M, 800 tested overlap pairs have median defect zero and cocycle fraction 1.0, against 0.801 for the shuffled control. These are finite structural receipts. Strict neutral bulk, Einstein-branch entry, physical particles, and physical CMB are open lanes in the gap register; the public-data rows above are separately classified paper-side comparisons [C7].

3. From closure coordinates to physical structure

The local pixel closure

The SL-3 estimate of the pixel from measured α is $P = 1.630968209403959 \dots$, a declared input of the claim lattice. A trial pixel P_k produces a field-width readback $R_P(P_k)$, which the pixel chart turns into the next iterate:

$$\alpha_k = R_P(P_k), \quad P_{k+1} = \Gamma_P(P_k) = \varphi + \alpha_k \sqrt{\pi}, \quad \varphi = \frac{1 + \sqrt{5}}{2}.$$

On the selected particle branch, the internal readback chain is

$$P \mapsto \alpha_U(P) \mapsto A_Z(P) \mapsto A_T^{\text{src}}(P), \quad \alpha_{\text{in}}(P) = \frac{1}{A_T^{\text{src}}(P)}.$$

Two declared maps carry certified fixed points [C1]. The gauge-width map has the certified self-consistent fixed point

$$\alpha^{-1} = 137.035660136946 \dots,$$

a relative 2.5×10^{-6} from the measured $\alpha^{-1} = 137.035999177(21)$ [S2] (ledger row CL-2). The source map has the certified root

$$\alpha_{\text{root}}^{-1} = 136.994835177413 \dots,$$

a relative 3.0×10^{-4} from measurement (CL-1), with the forward closure point $P_{\text{fwd}} = 1.630972095858897 \dots$ satisfying $P_{\text{fwd}} = \varphi + \sqrt{\pi}/A_T^{\text{src}}(P_{\text{fwd}})$. Both roots are enclosed by direct interval Banach with outward rounding, the enclosure covers the infinite-cutoff $\text{SU}(2)/\text{SU}(3)$ edge sums through geometric majorants, and the printed-pair identity holds to 35+ digits under converged precision-100 reruns, enforced by a CI test (CL-6). The certificates establish existence and uniqueness for the declared maps; the Ward-projected hadronic transport term of the physical Thomson readout is open, and its source-derived emission and same-scheme physical readout are work in progress. This lane has no registered target or falsification verdict.

The global capacity closure

Let Ω_N^{sc} be the closed-screen normal forms supported at capacity N . The global source map compares their record capacity with the screen budget and applies Minimal Admissible Realization (MAR):

$$S(N) = \log |\Omega_N^{\text{sc}}| - N, \quad N_\star = \text{MAR} \arg \max_N S(N).$$

The selected endpoint fixes the horizon normalization through

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}, \quad \Lambda_\star a_{\text{cell}} = \frac{3\pi P_{\text{fwd}}}{N_\star}, \quad a_{\text{cell}} = P_{\text{fwd}} \ell_\star^2.$$

The coupling theorem G2-GAP-1 is proven modulo three declared premises: the seed ($N_0 = \pi$), the contraction, and the tick-projection identity are theorems, the algebraic layer is machine-checked in Lean, and the conditional fixed point is certified at 3.532×10^{122} with relative width 1.6×10^{-25} [C16]. The estimate from measured Λ in the Planck base- Λ CDM model is approximately $N_\Lambda = 3.313 \times 10^{122}$ [S5]. The candidate bridge differs from the Λ -located capacity by 6.6%, so it is not the closure equation that forces the measured capacity; that equation is the object under search (ledger row CL-3). Construction of F and the premises CP-1–CP-3 are open; the balance condition CP-1 is the one counting-theorem-shaped premise between CL-7 and the live capacity test.

Read through $\Lambda_\star \ell_\star^2 = 3\pi/N$, the bridge is a zero-dial relation between the two constants that frame physics:

$$\Lambda \ell_P^2 = 3 \exp \left[-\frac{6\pi}{P \alpha_U(P)} \right] \quad (\text{up to the correction term}).$$

Since α_U is a function of P alone and P is located from α , the theory side holds no continuous freedom once α is measured. The 122 orders of magnitude that the naive vacuum-energy estimate misses appear as e^{-281} , and $281 = 6\pi/(P \alpha_U)$ is the same exponent that runs the weak hierarchy: one exponent carries both hierarchy problems. The equation that would force the measured capacity exactly is work in progress; the residual is structural (RG-truncation-shaped terms are excluded by scan). A Λ determination at percent-level precision would discriminate among completed candidate closures. This is a completion-stage comparison criterion, not a current falsification verdict.

The compact-gauge branch

The gauge derivation starts with individually transportable seeds selected by zero-obstruction transport. A common strict representative fixes the tensor category carried by those seeds. Surjective pullback functors and forgetful fibers construct its refinement limit, and Doplicher–Roberts/Tannaka reconstruction reads off the compact group. MAR selects the realized one-generation/one-Higgs matter package; anomaly cancellation, Yukawa invariance, and the central kernel then fix the hypercharge lattice and quotient [O4]. The result is

$$G_{\text{SM}} = \frac{\text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_Y}{\mathbb{Z}_6}, \quad N_c = 3, \quad N_g = 3.$$

The selected hypercharge lattice is

$$\begin{aligned} Q_i &= (3, 2)_{1/6}, & u_i^c &= (\bar{3}, 1)_{-2/3}, & d_i^c &= (\bar{3}, 1)_{1/3}, \\ L_i &= (1, 2)_{-1/2}, & e_i^c &= (1, 1)_1, & H &= (1, 2)_{1/2}, & i &= 1, 2, 3. \end{aligned}$$

The Higgs doublet is the Borel–Weil carrier

$$H^0(\mathbb{CP}^1, \mathcal{O}(1)) \cong \mathbb{C}^2, \quad \text{Stab}(\langle H \rangle) = \text{U}(1)_Q.$$

Its connected adjoint is $(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0)$, excluding mixed $(3, 2, \pm 5/6)$ generators. The $\mathbb{Z}_6$ line lattice has a minimum magnetic line of one electron-Dirac unit with required color magnetic charge and electromagnetic theta period 2π ; dynamical monopole existence is separate. The same branch supplies

$$\beta_{\text{EW}} = N_c + 1 = 4.$$

An exact bracket on the port coefficient space

Euler fixes only the total coordination charge 12. Twelve separate unit defects follow on a feasible integer branch with cost $h(k) \geq h(1)|k|$, strictly for $|k| \geq 2$. An inverse-port involution then gives six axes. For $P_i = u_i u_i^T$ and $Q_i = P_i - I_3/3$, maximizing the source-side tomography volume

$$\det \left(\sum_i P_i \right) \det \left(\sum_i |Q_i\rangle \langle Q_i| \right)$$

forces $\sum_i P_i = 2I_3$, $\sum_i |Q_i\rangle \langle Q_i| = \frac{4}{5}I_5$, and $(u_i \cdot u_j)^2 = 1/5$. The unique six-line configuration is the regular icosahedral axis system. A source-derived strictly completely monotonic pair cost supplies an alternative selector. Production of either selector is the receipt RP-A5.

On this branch the port coefficient module is $P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}$. Pairing antipodes, let $U = [u_1 \cdots u_6]$, $\Phi(b) = \sum_i b_i P_i$, $G = \text{im } U^T$, and $W = \ker U$. The outward faces orient W , with $\|\Omega_W\|^2 = 20 - 4\sqrt{5}$. The equivariant isomorphism

$$\Theta(b + d_G + d_W) = (\kappa_E(\tfrac{1}{2}Ud_G) + i\Phi(b), \kappa_W(d_W))$$

pulls back the block commutator and constructs

$$(P_{12}, [\cdot, \cdot]_\Theta) \cong \mathfrak{u}(3) \oplus \mathfrak{so}(3) = \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2).$$

Its center is the uniform even line, its derived algebra has dimension eleven, and $[iS, iT] = -2(E_{12} - E_{21}) \neq 0$ proves that the five-band is noncentral. This bracket acts on coefficient fluctuations. The central record projectors commute. A full-rank,

compact skew-adjoint, commutator-closed, refinement-natural physical current lift whose induced A_5 action is inner forces the same Lie type by compact Lie classification. Source production of that lift is work in progress.

The trace-balanced completion

$$\widehat{\Theta}(b + d_G + d_W) = \left(\kappa_E(\tfrac{1}{2}Ud_G) + i\Phi(b), \kappa_W(d_W) - \tfrac{i}{2}(\sum_i b_i)I_2 \right)$$

maps onto $\mathfrak{s}(\mathfrak{u}(3) \oplus \mathfrak{u}(2))$. Its connected integration is $S(U(3) \times U(2))$, and the cover $(A, B, z) \mapsto (z^2 A, z^{-3} B)$ has kernel $\mathbb{Z}_6$. Separately, $\Lambda_+ / (\Lambda_1 + \Lambda_5) \cong \mathbb{Z}_6$ with Smith invariants $(1, 1, 1, 1, 1, 6)$. The latter is an axis-load residue, not a rank-four cocharacter lattice. Physical identification requires determinant balance, a spin lift, implementer-sensitive loop readout, and refinement-natural descent. Primitive Screen Tomography plus Reversible Current (PSTRC) is one sufficient branch contract for these inputs; its production from the recovered core is work in progress.

Spherical multiplets restrict as $\ell = 2 : \mathbf{5}$, $\ell = 3 : \mathbf{3}' \oplus \mathbf{4}$, $\ell = 4 : \mathbf{4} \oplus \mathbf{5}$, $\ell = 5 : \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}$, and $\ell = 6 : \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{4} \oplus \mathbf{5}$. Every A_5 -invariant operator or ensemble covariance on $\ell = 2$ is scalar; one realization can be anisotropic. The equal-weight mask has $B_1 = \dots = B_5 = 0$ and $B_6/B_0 = 11/25$.

Status: conditional finite-carrier construction

Lean certifies the positive twelve-term sum-to-twelve unit split, abstract $\mathfrak{u}(3) \oplus \mathfrak{so}(3)$ commutator, dimension $9 + 3$, a matrix noncentrality witness, the six-axis lattice quotient, and the $11/25$ arithmetic. Euler charge, the cost, and the selector are not formalized. The port-frame map, A_5 -equivariance, compact-Lie classification, trace-balanced global form, and physical identification are paper proofs or open receipts. MAR is load-bearing for hypercharge, chiral content, and $N_g = 3$. The physical gates are UD12, RP-A5, PORT-CURRENT-INNER, BLOCK-DETERMINANT-BALANCE, PORT-SPIN-LIFT, and AXIS-CENTER-DESCENT. The source-to-observable transfer is work in progress.

The QCD-free hierarchy witness

The source-audit hierarchy witness reuses the pixel candidate and the compact-gauge integers. Its matched unified-width coordinate is $\alpha_U(P_{\text{fwd}}) = 0.041124247441816685$. The screen count begins with a triangulated S^2 . For curvature charge $q_v = 6 - \deg(v)$, Euler's identity and $3F = 2E$ give $\sum_v q_v = 6V - 2E = 12$. The UD12 and RP-A5 branch realizes twelve equivalent fivefold defects, and edge-center completion turns them into twelve central ports. Independently, reversible orientation doubles the product-adjoint dimension: $m_{\text{rep}} = 2 \dim(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)) = 2(8 + 3 + 1) = 24$. The oriented screen register also has 24 slots; equality of these counts does not make the screen selector a premise of the product-adjoint theorem. With local cell entropy $\ell_{\text{cell}} = P_{\text{fwd}}/4$ and $\beta_{\text{EW}} = 4$, the transmutation law is

$$\frac{v}{E_*} = P_{\text{fwd}}^{-1/2} \exp \left[-\frac{2\pi}{\beta_{\text{EW}} \alpha_U(P_{\text{fwd}})} \right] = 2.0198114078576331 \times 10^{-17},$$

evaluated at the certified P_{fwd} . The exact parametric law and the $12/24$ counts are theorem outputs; the numerical endpoint has the certificate class source-audit branch witness [O5,C3]. The weak scale is a settled-form readout with naturality defect $\epsilon_H = 0$, $\epsilon_H \in [0, 0]$: a seventeen-order hierarchy from a one-line law, with no supersymmetric machinery and no tuning.

The criticality branch and the electroweak chart

The criticality law $\lambda = 0$, $\beta_\lambda = 0$ at one source scale fixes the top Yukawa from the gauge couplings, a family with zero continuous parameters over the boundary scale; the boundary-scale selection is a variational theorem (exactly quadratic record cost on the one-loop chart, capacity-weighted log-mean placement) modulo two finite carrier facts, and the selected boundary is

$$\mu_b = \sqrt{\mu_U E_{\text{cell}}} = E_* e^{-\pi} P_{\text{fwd}}^{-1/6} = 4.86 \times 10^{17} \text{ GeV}.$$

Evaluated at the measured top mass, the fit-free criticality curve returns the Higgs coordinate 125.7 GeV at two loops, within 0.5% of the measured value [S4] and inside the declared truncation and matching bands (conditional criticality branch) [C4]. A flow-internal alternative is a certified no-go: $d\beta_\lambda/d \ln \mu$ has no root on $[10^{16}, 1.3 \times 10^{19}]$ GeV, so the boundary scale has to come from the observer-patch side, and the four-candidate boundary-scale registry is frozen before any three-loop computation exists [C4]. The declared Higgs/top surface arithmetic behind the electroweak chart is certified from raw interval data: a published interval input box for the eleven declared branch inputs, an outward-rounded enclosure of every formula node, a Jacobian interval enclosure over the full box, and a non-singular readout certificate, scoped to the declared surface [C3].

The strict source-audit branch also returns the running/tree chart coordinates $(m_W^{\text{chart}}, m_Z^{\text{chart}}) = (80.330, 91.119)$ GeV, with no measured mass entering; the physical readout contract (renormalized vev, tadpole scheme, thresholds, matching order, complex-pole map) is open, and no physical pull is defined. One convention diagnostic is on record: under the complex-pole convention with PDG 2026 central masses and widths, the converted W pole is 80.3340 GeV [S3], and the chart W coordinate sits 0.5 propagated experimental standard deviations from it. The diagnostic measures convention distance on the chart coordinate; a physical comparison requires the completed readout contract.

Strong-interaction scale

Dimensional transmutation of the source strong coupling, with no hadronic input, gives $\alpha_s(M_Z) = 0.11834$, 0.5σ from the public reference, and fixes the strong scale

$$\Lambda_{\text{QCD}}^{(3)} = 334.8 \text{ MeV},$$

0.3σ from the measured 338(12) MeV, on the conditional source-running branch. The published lattice ratio m_N/Λ_{QCD} places the nucleon at 929 MeV, -1.0% from measurement, as an external lattice input [C2,C7].

4. Geometric gravity normalization and the SI readout

On the declared Einstein branch, the edge-entropy/area-law theorem identifies the coefficient of the local Einstein equation

$$G_{ab} + \Lambda g_{ab} = 8\pi G_{\text{geom}} \langle T_{ab} \rangle$$

and its Newton–Poisson limit with the observer-cell ratio

$$G_{\text{geom}} = \frac{a_{\text{cell}}}{4\ell_{\text{shared}}}.$$

The local pixel branch supplies $a_{\text{cell}} = P_{\text{fwd}} \ell_\star^2$ and $\bar{\ell}_{\text{shared}} = P_{\text{fwd}}/4$. Substitution gives

$$\frac{G_{\text{geom}}}{\ell_\star^2} = \frac{P_{\text{fwd}}}{4(P_{\text{fwd}}/4)} = 1, \quad \boxed{G_{\text{geom}} = \ell_\star^2}.$$

The pixel number cancels because it counts both cell area and shared edge entropy. The factor 4 follows from the ratio between the structural Einstein coefficient and the Newton–Poisson convention [O4].

Newton coupling and the SI clock readout

A cesium clock translates the geometric identity into SI units:

$$\gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c} = \frac{\varepsilon_{\text{Cs}}}{2\pi}, \quad G_{\text{SI}} = \frac{c^3 \ell_\star^2}{\hbar} = \frac{c^5 \varepsilon_{\text{Cs}}^2}{4\pi^2 \hbar \nu_{\text{Cs}}^2}.$$

For the selected display coordinate $\gamma_\star = 4.955974336548419 \dots \times 10^{-34}$, this gives $G_{\text{SI}} = 6.674299995910528 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, beside the 2022 CODATA value $6.67430(15) \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ [S1]. The agreement is a unit-bookkeeping identity on the selected scale bridge (SL-5): the display coordinate $\gamma_\star$ is chosen on that bridge, so G enters as an input via the bridge. Evidence class: exact geometric normalization plus SL-5 bridge input display [O1,O4,C3].

5. Landmark solution map

The table states the positive result and records its evidence class once. Paper numbers refer to the numbered OPH papers in the References block.

Physics problem	OPH solution on the declared branch	Evidence class	Paper
Observer and measurement problem	An observer is a bounded self-reading record boundary. Compatible local readouts participate in the repair map that returns the public world $T(\mathfrak{U}) = \mathfrak{U}$.	Internal theorem	1,2
Quantum probabilities and update	The finite central record algebra yields the Born rule, Lüders conditioning, and $ S_{\text{CHSH}} \leq 2\sqrt{2}$ on the two-wing event surface.	Internal theorem	1,3
Time, Lorentz symmetry, and three space dimensions	Cap modular flow supplies the relative clock; $\text{Conf}^+(S^2) \cong \text{SO}^+(3,1)$, and $H^3 \cong \text{SO}^+(3,1)/\text{SO}(3)$ has dimension three over a $3+1$ event base.	Receipt-conditional theorem	4
General relativity	Null translations, local stress, generalized-entropy stationarity, and the uniform small-ball limit compose to $G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle$.	Receipt-conditional theorem	4
Gravity from quantum information	The edge-entropy/area law fixes the Einstein and Newton coefficient: $G_{\text{geom}} = a_{\text{cell}}/(4\bar{\ell}_{\text{shared}}) = \ell_\star^2$.	Internal identity	4
Standard Model gauge and matter structure	DR/Tannaka reconstruction plus MAR, anomaly cancellation, and Yukawa invariance give $(\text{SU}(3) \times \text{SU}(2) \times \text{U}(1))/\mathbb{Z}_6$, the hypercharge lattice, $N_c = N_g = 3$, and one Higgs doublet.	Realized-branch theorem	4
Why the gauge algebra is $8+3+1$	The oriented twelve-port coefficient module admits an explicit compact bracket of Standard-Model Lie type. Every full-rank compact commutator-closed, refinement-natural current lift carrying the same inner A_5 action has that type uniquely.	Exact construction; physical forcing conditional on UD12, RP-A5, and PORT-CURRENT-INNER	4
Why there are three generations	On the realized MAR branch, CKM phase counting, weak-sector asymptotic freedom, and minimal admissibility give $N_g = 3$. A separate geometric route observes that the smallest irreducible A_5 extension of a nontrivial oriented face- C_3 phase has dimension three. Connecting that carrier to the physical family fiber requires the open A5-FAMILY-ATTACHMENT receipt.	Realized-branch theorem by MAR; independent A_5 attachment open	4
Electroweak breaking and charge quantization	The Borel–Weil Higgs carrier $H^0(\mathbb{CP}^1, \mathcal{O}(1)) \cong \mathbb{C}^2$ has stabilizer $\text{U}(1)_Q$. The MAR matter package, anomaly/Yukawa constraints, and central tensor kernel fix the Standard Model charge lattice.	MAR-conditional theorem	4
Gauge-mediated proton-decay boundary	The connected product adjoint has no X/Y generator, so the standard simple-GUT channel is absent; general proton stability is not claimed.	Realized-branch corollary	1,4
Weak/UV hierarchy and Higgs naturality	The conditional 12-port screen alignment, product-adjoint $m_{\text{rep}} = 24$, P_{fwd} , and $\alpha_U(P_{\text{fwd}})$ emit $v/E_\star \sim 2.02 \times 10^{-17}$ with $\epsilon_H = 0$, without fixing a physical mass or coupling.	Conditional	5
Dark energy / cosmological term	Conditional on a constructed readback map F and CP-1–CP-3, the record-capacity branch gives $\Lambda_\star \ell_\star^2 = 3\pi/N_\star$, equivalently $\Lambda \ell_P^2 = 3e^{-6\pi/(P\alpha_U)}$; the 122 missing orders of the vacuum-energy problem as one exponential of the inverse gauge coupling, while vacuum energy lies in the kernel of the local null readout. No 10^{120} fine-tuning anywhere in the theory.	Conditional symbolic closure	1,4
Dark-matter phenomenology	The proposed repair-charge action has a dust-like normal phase and a cubic condensed phase with $a_R = \sqrt{a_b a_0}$ on the spherical deep branch. Its OPH derivation, constants, relativistic limit, and evidence are work in progress [S7].	Conditional action consequences	7
Yang–Mills gap	The transfer/repair theorem gives the conditional identity $\Delta_{\text{YM}} = \Delta_{\text{rep}}$ under compact-gauge continuum-branch assumptions that contain the Clay-hard content.	Conditional identity	8
Three distinct charged-lepton masses	Stratum theorem on the icosahedral carrier: the fivefold and threefold axes carry exactly doubly degenerate quadrupole spectra, so three distinct masses force the minimizing orbit off the high-symmetry strata; coefficient emission is an open lane.	Closed stratum theorem	5
String-vacuum selection	Observer-patch consistency defines a finite sieve over critical-string candidates; the Bouchard–Donagi geometry is selected on its declared gate stack.	Conditional selection program	9

What is distinctive about the program

OPH uses one finite observer-patch language for records, quantum event probabilities, overlap repair, emergent geometry, and transportable charges. Its exact finite claims, receipt-conditional physical branches, numerical map roots, and open physical maps are kept separate. Public code and Lean artifacts check the finite subset they implement. The closure ledger names every open map, while the falsification program is restricted to mature non-cosmological claims. This common reconstruction language and its explicit status boundaries are the main case for taking the program seriously.

6. The falsification program: how to kill this theory

OPH publishes exact failure conditions for mature non-cosmological claims [C14]. The program contains ten mathematical and five physical realized-branch falsifiers, with no locked-target registry. The compact list is:

Test	Outcome that kills the branch
Finite consensus	Two inequivalent terminal quotient normal forms from one source under the stated termination, diamond, boundary, and quotient hypotheses.
Collar recovery	Violation of the stated recovery bound by a family satisfying its uniform conditional-mixing and boundary-growth premises.
Lorentz–Einstein branch	Failure of the Lorentz action, commutators, or Einstein equation on a receipt-complete model satisfying the published hypotheses.
Compact gauge and SM quotient	Failure of compact-gauge reconstruction, hypercharge, anomaly, shared-center, color, generation, or one-Higgs calculations under their realized-branch premises.
Classical carriers	Wrong transverse mode count for a declared Maxwell, perturbative pure Yang–Mills, or pure Einstein–Hilbert quadratic action under its stated vacuum and gauge assumptions.
Physical realized branch	A confirmed charge-lattice outlier, non-three color multiplicity, fourth light chiral generation, required extra elementary Higgs multiplet, or incompatible low-energy gauge quotient under the same branch conditions.
Finite screen sieve	A carrier satisfying the finite sieve premises violates the stated port, reserve, phase, or quotient conclusion.

Cosmology, dark matter, quantitative mass/readout closures, black-hole continuations, Yang–Mills mass-gap completion, and engineering lanes are outside this program [C14,C17].

The assent structure is mechanical. A reader can inspect the six principles, the cited artifacts, the branch receipts, and the closure ledger [C17] separately. The certified fixed points cannot be moved within their declared maps, and no OPH surface may claim more than the ledger records. A completed physical identification requires the open source-to-observable maps to close without target leakage and with their own uniqueness certificates.

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Data and standards

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